

Welcome to Math 1365

"Proofs"

Rigorous Mathematics

Logic

Lesson 1 Definitions, Theorems, Proofs

Statement: declarative sentence
which has a definite truth value,
i.e. either true or false but
not both

Examples

a) The sum of 2 even integers is even.
IS statement

b) $1+1=3$
false statement

c) $x+3=5$
"open sentence" $\rightarrow$ statement
true if $x=2$
false otherwise

d) $\frac{x+y}{x-y}$
"not" a statement

does not declare anything
e) This sentence is false
NOT a statement
If true, then false

Vice versa
(self-referencing)
f) What is Goldbach's conjecture?
- NOT a statement
- does not declare
- a question

3 types of math statements
Definition : "precise meaning"
Theorem : proven statement
Proof : a collection of logical arguments to prove a theorem

Ex
→ Def An integer x is even if $x = 2n$ multiply
for some integer n

Thm 8 is an even integer.

Proof Since $8 = 2(4)$ and 4 is an integer,
(minimal)

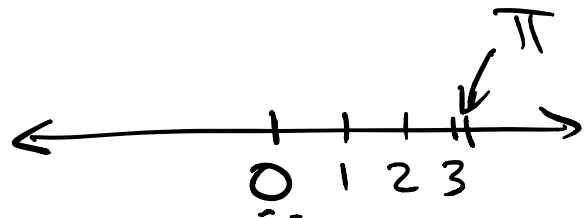
8 is even by def (of even integers).

"Writing" $\rightarrow$ art
 $\rightarrow$ proposition, result, fact $\rightarrow$ def
 $\rightarrow$ thm
 $\rightarrow$ axiom

- undefined terms
- unproven statement, we accept to be true (axiom)

1) Numbers

- real $\mathbb{R}$



- integers

positive, 0
+ve, -ve, 0

2) operations: $+$, $-$, $\times$, $\div$

$$\text{int} + \text{int} = \text{int}$$

3) properties of arithmetic

$$a + b = b + a \quad \text{commutative law}$$

$$(a + b) + c = a + (b + c)$$

$$a(b+c) = ab+ac$$

4) ordering of #'s

$$2 < 3, \quad 7/2 = 3.5$$

$$>, \leq, \geq$$

all -ve #'s $<$ positive #'s

many way to def

An integer x is even if
 $x/2$ is an integer

Proof: $8/2 = 4$ is an integer

Def An integer x is odd if
 $x = 2n + 1$ for some integer n

Thm 9 is odd

Proof $9 = 2(\underset{\substack{\uparrow \\ \text{integer}}}{4}) + 1$

Def An integer a divides an integer b , denoted $a|b$, if $b = an$ for some integer n .

(not a/b)

$$\underbrace{4}_a | \underbrace{12}_b \text{ because } 12 = 4 \underbrace{(3)}_{a(n)}$$

$$(12/4 = 3)$$

$$\underbrace{a|b}_{\dots}$$

$\longleftrightarrow$ a is a factor/
divisor of b
 $\longleftrightarrow$ b is multiple of a

"Divisibility" generalizes "even"
 $a=2$

$$\underbrace{a|b}_{\dots} \longleftrightarrow \underbrace{b}_{\dots} \text{ is } \underline{\underline{\text{divisible}}}_{\dots} \text{ by } \underbrace{a}_{\dots}$$

Def An integer $p > 1$ is prime if the only positive divisors

Of p are 1 and itself, p .

~~5~~ 1 and the only positive
divisors of 5 are 1 and 5
So 5 is prime.

2, 3, 4 are not divisors
of 5

$$\begin{array}{r} 2 \overline{) 5} \\ \underline{4} \\ 1 \end{array}$$

↑
Slash = not

$$5 \neq 2n \text{ for any integer } n$$

Def An integer $a > 0$ is
composite if it has a factor
 b s.t. $1 < b < a$

$$15 = 3(5)$$

3, 5 are factors of 15

$$1 < 3, 5 < 15$$

is composite.

Disprove = prove false

disprove $\overline{3}$ is even

prove 3 is not even

3 $\neq 2n$ for any integer n

not clear yet from def
odd $\rightarrow$ not even
 $2n+1$ $2n$